black hole are determined uniquely by its mass, electric charge, and spin.

nonlinear. The property of combining together in a more complicated way than simple addition.

nova. A brilliant outburst of light from an old star, now known to be caused by a nuclear explosion in the star's outer layers.

nuclear burning. Nuclear fusion reactions that keep stars hot and power hydrogen bombs.

nuclear force. Also called the "strong interaction." The force between protons and protons, protons and neutrons, and neutrons and neutrons, which holds atomic nuclei together. When the particles are somewhat far from each other, the nuclear force is attractive; when they are closer it becomes repulsive. The nuclear force is responsible for much of the pressure near the center of a neutron star.

nuclear reaction. The merging of several atomic nuclei to form a larger one (fusion), or the breakup of a larger one to form several smaller ones (fission).

nuclear reactor. A device in which a chain reaction of nuclear fissions is used to generate energy, produce plutonium, and in some cases produce electricity.

nucleon. Neutron or proton.

nucleus, atomic. The dense core of an atom. Atomic nuclei have positive electric charge, are made of neutrons and protons, and are held together by the nuclear force.

observer. A (usually hypothetical) person or being who makes a measurement.

old quantum mechanics. The early version of the laws of quantum mechanics, developed in the first two decades of the twentieth century.

optical astronomer. An astronomer who observes the Universe using visible light (light that can be seen by the human eye).

orbital period. The time it takes for one object, in orbit around another, to encircle its companion once.

paradigm. A set of tools that a community of scientists uses in its research on a given topic, and in communicating the results of its research to others.